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Inventor(s)	Baghestani; Saman et al.

System and method for generating financing structures using clustering

Abstract

Described herein is a system for generating financing structures. A learning engine may extract data sets associated with sellers of various products. The learning engine may be trained using the data sets. The learning engine may identify a subset of dimensions that cause a change in a determination of a final price for a given product. The learning engine may compute a value for each of the sellers with respect to each dimension. The learning engine may group the sellers into different clusters. The learning engine may generate using a model, including the subset of dimensions. The learning engine may receive a request to generate a financing structure for a specified product sold by a specified seller. The learning engine may generate financing structures for the specified product sold by the specified seller based on the generated model.

Inventors: **Baghestani; Saman (Plano, TX), Shah; Rohan (Frisco, TX), Dolle; Nicholas E. (Mckinney, TX)**

Applicant: **Capital One Services, LLC** (McLean, VA)

Family ID: **78512653**

Assignee: **Capital One Services, LLC (McLean, VA)**

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Primary Examiner: Rotaru; Octavian

Attorney, Agent or Firm: Sterne, Kessler, Goldstein & Fox P.L.L.C.

Background/Summary

BACKGROUND
(1) Different entities such as car dealerships, home builders, real estate agents, retail stores interface with different financial intuitions for financing a product for sale. The product may be automobiles, houses, appliances, furniture, and/or the like. Each of these entities may operate differently and may have different methodologies to determine a final sales price for a particular product.
(2) Determining a financing structure without understanding an entity's needs may be a slow and error-prone process. The financial institution may not generate financing structures that best fit the entity even after multiple iterations. Transmitting multiple requests and receiving multiple financing structures may be computationally expensive and create a network bottleneck.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) The accompanying drawings, which are incorporated herein and form part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles of the disclosure and enable a person skilled in the relevant art to make and use the disclosure.
(2) FIG. **1** is a block diagram illustrating data flow for a system for generating financing structures according to some embodiments.
(3) FIG. **2** is a block diagram illustrating clusters positioned on a multi-dimensional graph according to some embodiments.
(4) FIG. **3** is a block diagram illustrating various dimensions for the multi-dimensional graph according to some embodiments.
(5) FIG. **4** is a block diagram illustrating dimensions and weights for the multi-dimensional graph according to some embodiments.
(6) FIG. **5** illustrates a graph including nodes and edges, indicating a seller's score for different dimensions.
(7) FIG. **6** illustrates an example environment in which systems and/or methods for generating

financing structures are implemented according to some embodiments.
(8) FIG. **7** is a flowchart illustrating a process for authorizing secondary users to access a primary user's account, according to some embodiments.
(9) FIG. **8** is a block diagram of example components of a device, according to some embodiments.
(10) The drawing in which an element first appears is typically indicated by the leftmost digit or digits in the corresponding reference number. In the drawings, like reference numbers may indicate identical or functionally similar elements.

DETAILED DESCRIPTION

(11) Described herein is a system for generating financing structures. A learning engine may extract data sets associated with sellers of various products. Each data set may be associated with a dimension used for generating a financing structure. The learning engine may be trained using the data sets. The learning engine may identify a subset of dimensions that cause a change in a determination of a final price for a given product for more than a threshold amount of sellers and which cause a change in the determination of the final price more than a threshold amount. The learning engine may compute a value for each of the sellers with respect to each dimension of the subset of dimensions. The learning engine may group the sellers into different clusters. Each cluster may include a subset of sellers having a value for a given dimension within a predetermined amount of each other. The learning engine may be generated using a model including the subset of dimensions.
(12) The learning engine may receive a request to generate a financing structure for a specified product sold by a specified seller. The learning engine may generate financing structures for the specified product sold by the specified seller based on a generated model and an identified price of the specified product set by the specified seller. The learning engine may use a machine-learning algorithm to generate financing structures for the specified product sold by the specified seller. The machine learning algorithm may be an unsupervised machine learning algorithm, such as K-means clustering.
(13) The system provides a financing structure that may be predicted to be acceptable to the particular seller for a given product while meeting lender credit decisioning requirements and buyer preferences. This way, the seller and buyer are presented with financing structures that are specifically geared to effectively moving the parties to agreement on the financing structure enabling the product to be purchased given the pricing structure of the product. In this regard, the system solves the technical problem of transmitting multiple requests for generating financing structures and receiving each of the financing structures. In this configuration, the system reduces the amount of computer resources required and avoids potential network bottleneck.
(14) FIG. **1** is a block diagram illustrating data flow for a system for generating financing structures according to some embodiments. In operation **100**, a seller (e.g., automobile dealership) of a specified product (e.g., an automobile) may interface with a financial institution to generate a financing deal for purchasing the specified product.
(15) In operation **102**, the financing deal may go through an underwriting process. For example, the financial institution may underwrite the financing for purchasing the specified product. The underwriting process may generate a decision structure **109** for the financing deal. The decision structure may govern the rules of determining a final financing structure. For example, multiple different attributes may be factored into determining the final financing structure. The multiple different attributes may include the final sale price of the specified product, the breakdown of the sale price, the amount of down payment being provided, details related to the product (e.g., condition of the product, whether the product is new or used, type of product, and/or the like), and/or the like. The decision structure may vary for each financial institution and seller.
(16) The financing structure details may be stored in a data store **104**, after the underwriting process. The data store **104** may store information such as the financing structure details (e.g., APR, total amount financed, time period of the loan, customer details, credit score of the customer,

and/or the like). Data store **104** may also store information such as recalculations, callbacks, and contracts. Recalculations may be different calculations executed to arrive at the final financing structure. Callbacks may be counter offers offered by the financial institutions to arrive at the final financing structure. For example, a financial institution may approve a customer for certain financing offers; however, may not approve the customer for a specific financing structure. In this situation, the financial institution may present a counteroffer to the specific financing structure that was not approved by the financial institution. The financing structure may include a combination of the type of product and specific structure aspects (sales price, back end products, trade-in value if present, cash down, etc.). A financial institution may present counteroffers, including different financing structures, until the parties come to an agreement on the final financing structure. In this regard, the financing structure is optimized for the customer and seller. Data store **104** may store information related to financing structures generated for different sellers **110** of different products.

(17) A dealer profile engine **106** may continuously receive the information stored in data store **104** as well as any financing structure after the underwriting processing, including generated decision structure **109**. Dealer profile engine **106** may be trained using the data received from data store **104** and generated decision structures **109**. Dealer profile engine **106** may group various sellers into different clusters **110** based on the data received from data store **104** and generated decision structures **109**.

(18) Dealer profile engine **106** may determine which attributes impact the determination of the final financing structure for more than a threshold amount of sellers **110**. Furthermore, dealer profile engine **106** may determine which attributes impact the final financing structure more than a threshold amount. Dealer profile engine **106** may then compute a value for each seller for each attribute, based on the data received from data store **104** and generated decision structures **109**. Dealer profile engine **106** may group sellers **110** into clusters based on their value for each attribute. For example, a cluster may include sellers **110** with values for a given attribute within a threshold amount. Dealer profile engine **106** may generate a model such as a multi-dimensional graph. Each dimension of the multi-dimensional graph may correspond with an attribute determined to impact the determination of the final financing structure for more than a threshold amount of sellers **110** and impact the final financing structure more than a threshold amount. Dealer profile engine **106** may position the clusters on the multi-dimensional graph. Dealer profile engine **106** generates alternative financing structures **108** based on the position of the cluster(s), including the seller, on the multi-dimensional graph. The multi-dimensional graph will be described in greater detail with respect to FIG. **2**.

(19) The alternative financing structures **108** generated by the dealer profile engine **106** are generated based on a predictive model that may anticipate and facilitate an agreement between the customer and seller on an alternative financing structure **108**.

(20) FIG. **2** is a block diagram illustrating clusters positioned on a multi-dimensional graph according to some embodiments. Dealer profile engine **106** may receive various sets of data **208**. As stated above, dealer profile engine **106** may receive decision structures for generating financing structures and data from the data store. The data of the decision structures and the data from the data store may be included in sets of data **208**. Sets of data **208** may include a sales price, backend value of the sales price, frontend value of the sales price, percentage of the sales price that is the backend value as compared to the total value of the product, funded structured flag, and/or the like. Dealer profile engine **106** may be continuously trained using sets of data **208**. The funded structured flag may indicate whether the seller prefers protection for assuming the credit risk.

(21) As described with respect to FIG. **1**, dealer profile engine **106** may determine which attributes impact the determination of the final financing structure for more than a threshold amount of sellers and which attributes impact the final financing structure more than a threshold amount, resulting in a subset of attributes. Dealer profile engine **106** may generate a dimension for each attribute in the subset of attributes, resulting in a subset of dimensions.

(22) Dealer profile engine **106** may then compute a value for each seller for each dimension. Dealer profile engine **106** may group sellers into clusters based on their value for each attribute. Dealer profile engine **106** may generate a model such as a multi-dimensional graph **210**. Each dimension, x1, x2, xn, of multi-dimensional graph **210**, may correspond with a different attribute. Dealer profile engine **106** may position clusters **200-206** on the multi-dimensional graph based on the values of each of the sellers for a given attribute.
(23) As a non-limiting example, cluster **200** may include sellers which presented counter offers of the sales price more often than other sellers (e.g., countered on sales price). The dimension x2 may correspond with the frequency of counteroffers presented by sellers. Accordingly, dealer profile engine **106** may compute a higher value for dimension x2 for cluster **200** than clusters **202-206**. Dealer profile engine **106** may position sellers in cluster **200** with respect to their values with respect to dimension x1, x2, and xn.
(24) As another non-limiting example, cluster **202** may include sellers which presented offers with a sales price including a higher percentage of backend value as compared to total price of the product, compared to sellers in clusters **200**, **204**, **206** (e.g., high BE as a % of book value (BV)). Dimension x1 may correspond to a percentage of backend value as compared to a total price of the product. Accordingly, dealer profile engine **106** may compute a higher value for dimension x1 for sellers in cluster **202** than clusters **200**, **204**, **206**. Dealer profile engine **106** may position sellers in cluster **202** with respect to their values with respect to dimension x1, x2, and xn.
(25) As another non-limiting example, cluster **206** may include sellers who did not bargain on the sales price of the product (e.g., no-haggle dealers). Accordingly, as dimension x2 corresponds with the frequency of counteroffers presented by sellers, dealer profile engine **106** may compute a lower value for dimension x2 for cluster **204** than clusters **200**, **202**, **206**. Dealer profile engine **106** may position sellers in cluster **204** with respect to their values with respect to dimension x1, x2, and xn.
(26) Each cluster may be identified by an attribute. For example, cluster **200** may be identified as Cluster-1: Countered on Sales Price; cluster **202** may be identified as Cluster-2: High BE as a percentage of book value (BV); cluster **206** may be identified as Cluster-3: no-haggle dealers; and cluster **204** may be identified as cluster-k: k attribute.
(27) In an embodiment, attributes can also include behavioral tendencies of various sellers. Attributes may include percentages of vehicles of a given make and model, seasonal variability, amount of time a product has been on sale, financial bonuses or goals of the seller, and/or the like. As an example, an automobile dealership may sell more of a vehicle of a given type of make and model as they may have more inventory of the vehicles of the given make and model. The automobile dealership may receive the vehicles of the given make and model and a lower price for purchasing the vehicles in bulk. In light of this, the automobile dealer may be more flexible in negotiating the price of the vehicle on the given make and model. In another example, the automobile dealership may have certain financial goals at the end of key dates of the year (e.g., end of the quarter, end of the year, end of the fiscal year, etc.). The price of vehicles may be affected based on the time of the year in an effort for the automobile dealership to meet their respective goals. In yet another example, a vehicle may remain unsold for a large amount of days. The price of this vehicle may be affected based on an effort for the automobile dealership to sell this vehicle as soon as possible.
(28) FIG. **3** is a graph illustrating various dimensions for the multi-dimensional graph according to some embodiments. A graph **300** may include an x-axis **302** and a y-axis **304**. X-axis **302** may include different types of dimensions and values of the dimensions as they impact a final sales price or financing structure of a product for various sellers. Y-axis **304** also includes different types of dimensions and values of the dimensions as they impact a final sales price or financing structure of a product for various sellers.
(29) As described above, dealer profile engine **106** may position clusters of sellers on a multi-dimensional graph based on the values of each of the sellers for a given attribute. The position of

the seller on the multi-dimensional graph may determine a seller profile. The seller profile may indicate patterns of seller behavior in determining the final price of a product.

(30) The x-axis **302** and y-axis **304** may represent n-dimensions of attributes that may be used to build the seller profiles. The distribution of each of the graphs may indicate how statistically reliable each of the n-dimensions of a seller profile are in determining how to characterize any given seller. For example, x-axis **302** may represent various scenarios such as counter offers by dollar amount intensity, and the y-axis **304** may represent credit tiers, income of the applicant, or the net trade-in value, and/or the like. The graphs can identify predictable patterns of seller behavior based on the x-axis **302** and y-axis **304**. For example, graphs can identify predictable patterns of seller behavior in response to any given combination of product/customer/structure scenarios to identify patterns and assess the strength of those patterns.

(31) As described above, the dealer profile engine may determine which attributes impact the determining of the sales price or final financing structure which is identified as impacting the final financing structure of sales price more than a threshold amount of occurrences and which attributes impact the sales price or final financing structure more than a threshold amount, resulting in a subset of attributes. Each attribute can correspond with a different dimension in the multi-dimensional graph.

(32) The dealer profile engine may determine that an attribute impacts the sale price or final financing structure for more than a threshold amount of sellers by determining that the attribute changes the determination or decision structure for a given seller in determining the sale price or final financing structure. The dealer profile engine may determine that the attribute impacts the sale price or final financing structure for more than a threshold amount of sellers based on the generation of a bell curve **306** at a specified value of the attribute.

(33) In a non-limiting example, the seller may be an automobile dealership, and a dimension may be a frequency of counteroffers presented to a buyer. The dealer profile engine may determine that the sales price or final financing structure presented by more than a threshold amount of automobile dealers was affected for automobiles over $30,000. Furthermore, the dealer profile engine may determine that the difference in sales price or final financing structure presented by the more than threshold amount of dealers varied more than a threshold amount based on their willingness to present counteroffers. As an example, automobile dealer A may have refused to present counteroffers for automobiles over $30,000, as compared to automobile dealer B, which frequently presented counter offers for automobiles over $30,000, resulting in a difference in the final sales price for the same automobile of $5,000. In the event the threshold amount is any amount more than $2,000, the dealer profile engine may determine that frequency of presenting counter offers is an impactful attribute.

(34) As another example, a dimension may be the weather on the day of a sale. The dealer profile engine may determine that the sales price or final financing structure presented by less than a threshold amount of automobile dealers was affected by the weather. Furthermore, for those automobile dealers for which the weather did affect the sales price or final financing structure, the amount the sales price or final financing structure was less than a threshold amount. In this case, the dealer profile engine may not deem the weather the day of the sale as an impactful attribute.

(35) FIG. **4** is a chart **400** illustrating dimensions and weights for the multi-dimensional graph according to some embodiments. Chart **400** includes dimensions **402** along the x-axis and feature weights **404** along the y-axis. Each dimension may be assigned a different weight based on how much the dimension impacts the sales price or final financing structure of a product. The individual weights within each of the dimensions (in this example, there are six dimensions) represent different observed values for attributes such as customer income or product information. The weights may be coefficients that establish how much weight to give to each of those attributes as the system scores a particular financing structure along each dimension.

(36) The dimensions represent the n-number of facets of differentiation (e.g., fixed price or not,

Backend utilization, LTV utilization, etc.). Within each facet of differentiation, the weights individually establish scoring parameters that contribute to scoring each final price or final financing structure against an established seller profile. The values assigned for explained variance represent the delta between expected and observed outcomes. The dealer profile engine may recalibrate the weighting of attributes as it receives more data. In this way, the variances function to inform continuous improvement in the weighting of the individual attributes within each dimension such that the variance between observed and expected outcomes becomes smaller and, in-turn, more accurate, over time.

(37) In one embodiment, if a dimension is impactful in the determination of the sales price or final financing structure of a product, then the weight for that dimension is higher. Conversely, if a dimension is not impactful in the determination of the sales price or final financing structure of a product, then the weight of the dimension is lower.

(38) In some embodiments, one dimension can affect the impact of another dimension. For example, if the product for sale is an automobile, a first dimension is the year of the automobile, and a second dimension is the color of the automobile. The color of the automobile may matter more for determining the final sales price when the automobile is less than ten years old. Conversely, the color of the automobile may not matter as much for determining the final sales price when the automobile is more than ten years old. In this regard, the impact of the second dimension of color may vary depending on the value of the first dimension.

(39) FIG. **5** illustrates a graph **500** including nodes and edges, indicating a seller's score for different dimensions. Graph **500** includes dimension lines **502** with maximum nodes **503** connected by edges **504**. The maximum nodes **503** may indicate a maximum value for each dimension along dimension line **502**. The maximum value may indicate that the dimension is the most impactful in determining a sales price or final financing structure for a product.

(40) As described above, the dealer profile engine may determine a value for each seller for each dimension. The value may correspond to how impactful that particular dimension is for the seller in determining the sales price or final financing structure. For example, the value may indicate how much the sales price or final financing structure changes based on a given dimension.

(41) The calculated values for the seller may be plotted on the graph **500** along dimension lines **502**. The values may be represented by nodes **508**. Nodes **508** may be connected by edges **506**. Nodes **508** may be connected to identify values along dimension lines **502** of a single seller. The closer nodes **508** are to maximum nodes **503** along the dimension lines **502**, the higher the calculated value for the seller for that particular dimension. In this regard, the higher the value (and the closer the node **508** is to maximum node **503**), the more impactful that particular dimension is for determining sales price or final financing structure for a product.

(42) FIG. **6** illustrates an example environment in which systems and/or methods for generating financing structures are implemented according to some embodiments. The environment may include a dealer profile engine **106**, a data store **104**, a user device **646**, external data sources **660**, and a financial institution device **670**. Dealer profile engine **160** may include a modeling engine **600**, a dimension engine **602**, an output engine **604**, and an extraction engine **606**. User device **646** may include an application **648** and a user interface **650**. User device **646** may use application **648** to interface with dealer profile engine **106** and financial institution device **670**. Financial institution device **670** may include an application **672** and a user interface **674**. The application **672** may be used to interface with dealer profile engine **106** and user device **646**. Dealer profile engine **106** may be embodied as a learning engine. Application **648** may interface with dealer profile engine **106** to render financing structures on user interface **650**. Data store **104** may store information regarding product sales by various sellers including, the final sales price of the product, details regarding the product, financing structures presented to the customer, final financing structure accepted by the customer, contracts, and/or the like. External data sources **660** may store information such as seller data including product inventory, decision structures used by the seller to determine the sales price

of a product, average sales prices provided by the seller, negotiation characteristics, and/or the like. External data sources **660** may include but are not limited to, websites, databases, data repositories, excel files, and/or the like.

(43) The devices of the environment **600** may be connected through wired connections, wireless connections, or a combination of wired and wireless connections. Dealer profile engine **106** and data store **104** may reside inside cloud computing environment **640**. Alternatively, Dealer profile engine **106** and data store **104** may reside outside cloud computing environment **640**.

(44) In an example embodiment, one or more portions of the network **630** may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless wide area network (WWAN), a metropolitan area network (MAN), a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a cellular telephone network, a wireless network, a WiFi network, a WiMax network, any other type of network, or a combination of two or more such networks.

(45) Backend platform **625** may include a server or a group of servers. In an embodiment, backend platform **625** may be hosted in a cloud computing environment **640**. It may be appreciated that backend platform **625** may not be cloud-based, or may be partially cloud-based.

(46) Cloud computing environment **640** may include an environment that delivers computing as a service, shared resources, services, etc. Cloud computing environment **640** may provide computation, software, data access, storage, and/or other services that do not require end-user knowledge of a physical location and configuration of a system and/or a device that delivers the services. Cloud computing system **640** may include computer resources **626**.

(47) Each computing resource **626***a-d* may include one or more personal computers, workstations, computers, server devices, or other types of computation and/or communication devices. Computing resource(s) **626***a-d* may host backend platform **625**. The cloud resources may include compute instances executing in cloud computing resources **626***a-d*. Cloud computing resources **626***a-d* may communicate with other cloud computing resources **626***a-d* via wired connections, wireless connections, or a combination of wired or wireless connections.

(48) Computing resources **626***a-d* may include a group of cloud resources, such as one or more applications (“APPs”) **626-1**, one or more virtual machines (“VMs”) **626-2**, virtualized storage (“VS”) **626-3**, and one or more hypervisors (“HYPs”) **626-4**.

(49) Application **626-1** may include one or more software applications that may be provided to or accessed by user device **646**. In an embodiment, application **648** may execute locally on user device **646**. Alternatively, the application **626-1** may eliminate a need to install and execute software applications on user device **646**. The application **626-1** may include software associated with backend platform **625** and/or any other software configured to be provided across the cloud computing environment **640**. The application **626-1** may send/receive information from one or more other applications **626-1**, via the virtual machine **626-2**.

(50) Virtual machine **626-2** may include a software implementation of a machine (e.g., a computer) that executes programs like a physical machine. Virtual machine **626-2** may be either a system virtual machine or a process virtual machine, depending upon the use and degree of correspondence to any real machine by virtual machine **626-2**. A system virtual machine may provide a complete system platform that supports the execution of a complete operating system (OS). A process virtual machine may execute a single program and may support a single process. The virtual machine **626-2** may execute on behalf of a user (e.g., user device **646**) and/or on behalf of one or more other backend platforms **625**, and may manage the infrastructure of cloud computing environment **640**, such as data management, synchronization, or long-duration data transfers.

(51) Virtualized storage **626-3** may include one or more storage systems and/or one or more devices that use virtualization techniques within the storage systems or devices of computing resources **626***a-d*. With respect to a storage system, types of virtualizations may include block

virtualization and file virtualization. Block virtualization may refer to abstraction (or separation) of logical storage from physical storage so that the storage system may be accessed without regard to physical storage or heterogeneous structure. The separation may permit administrators of the storage system flexibility in how administrators manage storage for end users. File virtualization may eliminate dependencies between data accessed at a file-level and location where files are physically stored. This may enable optimization of storage use, server consolidation, and/or performance of non-disruptive file migrations.
(52) Hypervisor **626-4** may provide hardware virtualization techniques that allow multiple operations systems (e.g., "guest operating systems") to execute concurrently on a host computer, such as computing resources **626***a-d*. Hypervisor **626-4** may present a virtual operating platform to the guest operating systems and may manage the execution of the guest operating systems multiple instances of a variety of operating systems and may share virtualized hardware resources.
(53) In an embodiment, dealer profile engine **106** may receive data from data store **104** and may extract data from external data sources **660**. The dealer profile engine **106** may continuously be trained using the data from data store **104** and external data sources **660**. As described above, data store **104** may store information regarding product sales by various sellers including, the final sales price of the product, details regarding the product, financing structures presented to the customer, final financing structure accepted by the customer, contracts, and/or the like. External data sources **660** may store information such as seller data including product inventory, decision structures used by the seller to determine the sales price of a product, average sales prices provided by the seller, negotiation characteristics, and/or the like.
(54) Dimension engine **602** may identify attributes that impact the determination of a sales price or final financing structure presented to a customer for a given product. Dimension engine **602** may identify a subset of attributes from multiple different attributes that cause a change in a determination of a final price for a given product identified more than a threshold amount of occurrences and which cause a change in the determination of the final price more than a threshold amount. Dimension engine **602** may eliminate the attributes based on the data from data store **104** and external data sources **660**. Dimension engine **603** may remove data received from data store **104** or external data sources **660**, which pertain to the eliminated attributes. Dimension engine **602** may transform the subset of attributes into a subset of dimensions for generating a multi-dimensional graph.
(55) Modeling engine **600** may compute a value for each seller with respect to each dimension of the subset of dimensions. The higher the value for the seller for a given dimension, this indicates the more impactful that particular dimension is in determining a sales price or final financing structure for that seller. Modeling engine **600** may group each of the sellers into different clusters. Each cluster may include a subset of sellers having a value for a given dimension within a predetermined amount of each other. Modeling engine **600** may generate a model, including the subset of dimensions. The model may be a multi-dimensional graph. The dimensions of the multi-dimensional graph may be defined by the subset of dimensions. Modeling engine **600** may position each cluster on a multi-dimensional graph based on each value of each seller in each cluster for a given dimension.
(56) In some embodiments, a given seller of a given product may generate a pricing structure for the given product being purchased by a customer. Dealer profile engine **106** may receive a request to determine a financing structure for the given seller and product. Output engine **604** may determine a cluster, including the given seller. Output engine **604** may determine the values for each dimension for the given seller. Output engine **604** may generate a financing structure for the given seller and product based on the position of the identified cluster, including the given seller on the multi-dimensional graph and the values for each dimension for the given seller. Output engine **604** may generate a financing structure that is optimized for the customer, given the pricing structure of the product. For example, the financing structure may provide the lowest APR,

optimized payment schedule, lowest monthly payment, and/or the like for the given customer, product, and pricing structure. In an embodiment, the request may include a final sales price for the product.
(57) In another embodiment, output engine **604** may forecast a final sales price for the given product sold by the given seller based on the values for each dimension for the given seller. In an embodiment, output engine **604** may generate multiple different alternative financing structures. The financing structure may include annual percentage rate (APR), the period of the loan, amount of loan, date of origination of the loan, customer details, and/or the like.
(58) In an embodiment, user device **646** may transmit the request for generating the financing structure using application **648**. Output engine **604** may transmit the financing structure to user device **646**. Application **648** may receive the financing structure and render financing structure on user interface **650**.
(59) Each time a financing structure is generated for a seller and the seller (or customer) accepts or rejects the financing structure, the data related to the financing structure is stored in the data store **104**. As stated above, the dealer profile engine **106** may continuously be trained using the data in the data store.
(60) The dealer profile engine **106** may use a machine-learning algorithm to generate financing structures for the specified product sold by the specified seller. The machine learning algorithm may be an unsupervised machine learning algorithm, such as K-means clustering.
(61) K-means clustering is a type of unsupervised learning algorithm. The algorithm may identify groups (or clusters) in the data. The number of groups may be identified by the variable K. The algorithm works iteratively to assign each data point to one of the K groups based on the features that are provided. Using K-means clustering, the dealer profile engine **106** may attempt to group a seller into a cluster based on their values for each dimension.
(62) As a non-limiting example, dealer profile engine **106** may be used to generate financing structures for automobile sales. Dealer profile engine **106** may receive data from data store **104** and may extract data from external data sources **60**. The dealer profile engine **106** may continuously be trained using the data from data store **104** and external data sources **660**. Data store **104** may store information regarding automobile sales by various dealers including, the final sales price of the automobiles, details regarding the automobiles, financing structures presented to the customer, final financing structure accepted by the customer, contracts, and/or the like. External data sources **660** may store information such as dealer data including automobile inventory, decision structures used by the seller to determine the sales price of an automobile, average sales prices provided by the automobile, negotiation characteristics, and/or the like.
(63) Dimension engine **602** may identify attributes that impact the determination of a sales price or final financing structure presented to a customer for a given product. Dimension engine **602** may identify a subset of attributes from multiple different attributes, which cause a change in a determination of a final price for a given product for more than a threshold amount of sellers and which causes a change in the determination of the final price more than a threshold amount. Dimension engine **602** may eliminate the attributes based on the data from data store **104** and external data sources **660**.
(64) As an example, a dealer may factor in automobile make and model, year, color, condition, time of the month, inventory, demand, fuel type, and/or the like in determining a sales price for the automobile. Moreover, certain dealers may be willing to negotiate more than others. Some dealers may be willing to negotiate on the total value of the automobile by increasing the backend value of services. For example, backend services may include extended warranties and service of the vehicle. Some dealers may reduce the total value of the automobile by adding in the backend value to the final sales price. In this regard, an attribute may include a percentage of backend value and/or front end value of the sales price. Out of all of these attributes, some attributes may affect or change the sales prices more than a threshold amount as compared to the other attributes.

Additionally, some attributes may affect or change the sales price for more than a threshold amount of dealers. Dimension engine **601** may transform the subset of attributes into a subset of dimensions for generating a multi-dimensional graph.

(65) Modeling engine **600** may compute a value for each dealer with respect to each dimension of the subset of dimensions. The higher the value for the seller for a given dimension, this indicates the more impactful that particular dimension is in determining a sales price or final financing structure for that seller. For example, the sales price for a given automobile may change more based on a dimension related to the percentage of backend value in the sales price for a first and second dealer as compared to a third and fourth dealer. Therefore, the value for that particular dimension may be higher for the first and second dealer as compared to the value for that particular dimension for the third and fourth dealer.

(66) Modeling engine **600** may group each of the dealers into different clusters. Each cluster may include a subset of dealers having a value for a given dimension within a predetermined amount of each other. Based on the above example, the first and second dealers may be grouped in a first cluster, and the third and fourth dealer may be grouped in a second cluster. Modeling engine **600** may generate a model, including the subset of dimensions. The model may be a multi-dimensional graph. The dimensions of the multi-dimensional graph may be defined by the subset of dimensions. Modeling engine **600** may position each cluster on a multi-dimensional graph based on each value of each dealer in each cluster for a given dimension.

(67) Modeling engine **600** may use the clusters to define dimensions of the multi-dimensional graph. For example, the modeling engine **600** may identify aggregate practices across all platforms. Modeling engine **600** may determine dimensions which are more relevant/important to a given cluster based on its position on the multi-dimensional graph. Modeling engine **600** may use the relevant/important dimensions for the given cluster to generate a multi-dimensional graph for finalized sales and financing structures for a given dealer. Modeling engine **600** may determine values for attributes associated with the relevant/important dimensions for each finalized sale or financing structure for the given dealer. Modeling engine **600** may group each finalized sale or financing structure in a cluster based on the values for the attributes for each finalized sale or financing structure.

(68) In some embodiments, dealer profile engine **106** may receive a request to determine a financing structure for a given dealer and automobile. Output engine **604** may determine the values of attributes for the requested financing structure based on the information provided in the request. Output engine **604** may determine a cluster of finalized financing structures that closely matches the values of attributes for the requested financing structure. Output engine **604** may generate a financing structure for the given dealer and automobile based on the identified cluster of finalized financing structures. The financing structure may be optimized for the customer based on the customer's preferences and based on a likelihood of the dealer agreeing to the finalized financing structures (e.g., based on the dealer's profile). For example, the customer may choose to purchase an extended warranty with the automobile. The output engine **604** will generate a financing structure that includes financing for the extended warranty in the pricing structure. This way, the dealer and customer are presented with financing structures that are specifically geared to the automobile being purchased, given the pricing structure of the automobile.

(69) In an embodiment, the user device **646** may be operated by the dealer. The dealer may transmit the request for generating the financing structure to the dealer profile engine **106**, using application **648**. Output engine **604** may transmit the financing structure to user device **646**. Application **648** may receive the financing structure and render financing structure on user interface **650**.

(70) In an embodiment, the dealer may transmit a request for generating a set of financing structures for a given automobile to the financial institution device **670** using application **648**. Application **672** may receive the request. Application **672** may interface with the dealer profile engine **106** to generate the set of financing structures. Application **672** may receive the set of

financing structures and transmit the set of financing structures to the user device **646**. Application **648** may receive the set of financing structures and render the set of financing structures on the user interface **650**.
(71) In an embodiment, the dealer may transmit a request for generating a financing structure for a given automobile (or a plurality of automobiles) to the financial institution device **670** using the application **648**. Application **672** may receive the request. Application **672** may interface with the dealer profile engine **106** to generate the financing structure for the given automobile or automobiles. Application **672** may receive the financing structure and transmit the financing structure to the user device **646**. Application **648** may receive the financing structure and render the financing structure on the user interface **650**.
(72) The financial institution may reject a specific financing structure for any given automobile and customer. The dealer may only allow for a single counteroffer to the specific financing structure. Application **672** may transmit a request for generating a counter offer financing structure to the dealer profile engine **106**. The dealer profile engine **106** may identify and generate a counteroffer financing structure that best fits the dealer's tendencies and requirements, as described above. Application **672** may receive the counter offer financing structure and transmit the financing structure to the user device **646**. Application **648** may receive the counter offer financing structure and render the counter offer financing structure on the user interface **650**. In an embodiment, dealer profile engine **106** may generate multiple different counteroffer financing structures and select the one that best fits the particular dealer.
(73) Each time a financing structure is generated for the seller (or customer) and the seller (or customer) accepts or rejects the financing structure, the data related to the financing structure is stored in the data store **104**. As stated above, the dealer profile engine **106** may continuously be trained using the data in the data store **104**. For example, the dealer profile engine **106** may identify attributes associated with the dealer's determination of the final price based on the new data that is collected, which may change the dealer's values with respect to one or more dimensions. In this regard, the dealer profile engine **106** is in near real-time or at least periodically (e.g., daily, hourly, weekly, monthly, etc.) being optimized based on new data collected to generate accurate financing structure geared to effectively moving the parties to agreement on the financing structure enabling the product to be purchased given the pricing structure of the product.
(74) FIG. **7** is a flowchart illustrating a process for generating financing structures, according to some embodiments. It is to be appreciated the operations may occur in a different order, and some operations may not be performed.
(75) Flowchart **700** starts at operation **702**. In operation **702**, an extraction engine may extract data sets associated with sellers. Each data set may be associated with a dimension used for generating a financing structure. The extraction engine may receive data from a data store or may extract data from external data sources.
(76) In operation **704**, a dealer profile engine may be trained using the data sets. The dealer profile engine may be trained to identify a subset of dimensions using a dimension engine, generate a multi-dimensional graph using a modeling engine, and generate and output financial structures using the output engine. The dealer profile engine is continuously trained using the plurality of data sets.
(77) In operation **706**, the modeling engine may identify a subset of dimensions that the plurality of sellers use in a determination of a final price for a given product for more than a threshold amount of occurrences.
(78) In operation **708**, the modeling engine may compute a value for each of the plurality of sellers with respect to each dimension of the subset of dimensions.
(79) In operation **710**, the modeling engine may group the sellers into different clusters. Each cluster including a subset of sellers having a value for a given dimension within a predetermined amount of each other.

(80) In operation **712**, the modeling engine may generate a model including the subset of dimensions based on the plurality of clusters. The model may be a multi-dimensional graph. The modeling engine may position the clusters along each of the subsets of dimensions on a graph, based on each value for a given dimension of each of the sellers in a cluster.
(81) In operation **714**, the output engine may generate financing structures for a specified product sold by a seller based on the generated model and a final price of the specified product set by the seller. The output engine may generate the financing structure for a seller by determining the position of a cluster, including the seller along each of the subset of dimensions. The output engine may receive a request for generating the financing structures based on a request received from a user device or a financial institution device.
(82) FIG. **8** is a block diagram of example components of device **600**. One or more computer systems **800** may be used, for example, to implement any of the embodiments discussed herein, as well as combinations and sub-combinations thereof. Computer system **800** may include one or more processors (also called central processing units, or CPUs), such as a processor **804**. Processor **804** may be connected to a communication infrastructure or bus **806**.
(83) Computer system **800** may also include user input/output device(s) **803**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **806** through user input/output interface(s) **802**.
(84) One or more of processors **804** may be a graphics processing unit (GPU). In an embodiment, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.
(85) Computer system **800** may also include a main or primary memory **808**, such as random access memory (RAM). Main memory **808** may include one or more levels of cache. Main memory **808** may have stored therein control logic (i.e., computer software) and/or data.
(86) Computer system **800** may also include one or more secondary storage devices or memory **810**. Secondary memory **810** may include, for example, a hard disk drive **812** and/or a removable storage device or drive **814**.
(87) Removable storage drive **814** may interact with a removable storage unit **818**. Removable storage unit **818** may include a computer-usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **818** may be program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface. Removable storage drive **814** may read from and/or write to removable storage unit **818**.
(88) Secondary memory **810** may include other means, devices, components, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **800**. Such means, devices, components, instrumentalities, or other approaches may include, for example, a removable storage unit **822** and an interface **820**. Examples of the removable storage unit **822** and the interface **820** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.
(89) Computer system **800** may further include a communication or network interface **824**. Communication interface **824** may enable computer system **800** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and collectively referenced by reference number **828**). For example, communication interface **824** may allow computer system **800** to communicate with external or remote devices **828** over

communications path **826**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **800** via communication path **826**.

(90) Computer system **800** may also be any of a personal digital assistant (PDA), desktop workstation, laptop or notebook computer, netbook, tablet, smartphone, smartwatch or other wearables, appliance, part of the Internet-of-Things, and/or embedded system, to name a few non-limiting examples, or any combination thereof.

(91) Computer system **800** may be a client or server, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software ("on-premise" cloud-based solutions); "as a service" models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), desktop as a service (DaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms.

(92) Any applicable data structures, file formats, and schemas in computer system **800** may be derived from standards including but not limited to JavaScript Object Notation (JSON), Extensible Markup Language (XML), Yet Another Markup Language (YAML), Extensible Hypertext Markup Language (XHTML), Wireless Markup Language (WML), MessagePack, XML User Interface Language (XUL), or any other functionally similar representations alone or in combination. Alternatively, proprietary data structures, formats, or schemas may be used, either exclusively or in combination with known or open standards.

(93) In some embodiments, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **800**, main memory **808**, secondary memory **810**, and removable storage units **818** and **822**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **800**), may cause such data processing devices to operate as described herein.

(94) It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

(95) The present disclosure has been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries may be defined so long as the specified functions and relationships thereof are appropriately performed.

(96) The foregoing description of the specific embodiments will so fully reveal the general nature of the disclosure that others may, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

(97) The breadth and scope of the present disclosure should not be limited by any of the above-

described exemplary embodiments but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A computer implemented method for increasing accuracy of weighing a plurality of attributes in a decision model, and revealing which dimension is most impactful in the decision model, the computer implemented method comprising: concurrently executing, by a plurality of hypervisors, multiple instances of a plurality of operating systems to provide hardware virtualization; performing block virtualization, by a virtualized storage component of the hardware virtualization, that separates logical storage from physical storage to allow attributes of a data store to be accessed without regard to a heterogeneous structure; receiving, at a profile engine, the attributes accessed from the data store; performing file virtualization, using a virtualized storage component of the hardware virtualization, to eliminate dependencies between the accessed attributes at a file-level and a location where files are physically stored; generating, by the profile engine, a multi-dimensional graph of a plurality of n-dimensions where each dimension of the n-dimensions corresponds with a respective attribute of the attributes; generating, by the profile engine, a bell curve along the n-dimensions of the multi-dimensional graph; determining, by the profile engine, which ones of the attributes impact the decision model more than a threshold amount according to a comparison of the attributes against the generated bell curve at a specified value for each of the attributes; identifying, by unsupervised machine learning, clusters of variable K to each of the attributes processed by K-means clustering that assigns each of the clusters to each of a plurality of K groups based on feature weights; positioning, by the profile engine on the multi-dimensional graph, the identified clusters, along a first axis comprising the n-dimensions and, along a second axis representing the feature weights, based on how much weight to give to each of the attributes along each of the n-dimensions; computing, by the profile engine, a distribution of each graph of the multi-dimensional graph along the first and second axes indicating how statistically reliable each of the n-dimensions are in determining predictable patterns of characterization for each of the identified clusters in response to respective combinations of the n-dimensions; recalibrating, by the profile engine, the feature weights of the attributes for the decision model, as the profile engine receives more data for the attributes according to a continuous decrease in variance between observed and expected outcomes of the decision model to improve accuracy over time of the weighing of each of the attributes within each of the n-dimensions; and generating, by the profile engine, along dimension lines of the multi-dimensional graph, a plurality of maximum nodes connected to a plurality of edges of a maximum value for each dimension of the n-dimensions along the dimension lines, indicating which particular dimension of the n-dimensions is most impactful in the decision model, according to said particular dimension having a higher value of its node up to the maximum nodes.
2. The computer implemented method of claim 1, wherein the decision model corresponds to generating a financing structure.
3. The computer implemented method of claim 1, where the attributes include: a final sale price of a specified product, a breakdown of a sale price, an amount of down payment being provided, condition of the specified product, whether the specified product is new or used, and a type of the specified product.
4. The computer implemented method of claim 1, wherein the n-dimensions include: a percentage of backend value as compared to a total price of a specified product, frequency of counteroffers presented by sellers, weather on a day of a sale, an age of the specified product, and a color of the specific product.
5. The computer implemented method of claim 1, wherein the profile engine is a dealer profile engine.

6. A system for increasing accuracy of weighing a plurality of attributes in a decision model, and revealing which dimension is most impactful in the decision model, the system comprising: a memory storing instructions; a processor in communication with the memory to execute the instructions to allow: concurrently executing, by a plurality of hypervisors, multiple instances of a plurality of operating systems to provide hardware virtualization; performing block virtualization, by a virtualized storage component of the hardware virtualization, that separates logical storage from physical storage to allow attributes of a data store to be accessed without regard to a heterogeneous structure; receiving, at a profile engine, the attributes accessed from the data store; performing file virtualization, using a virtualized storage component of the hardware virtualization, to eliminate dependencies between the accessed attributes at a file-level and a location where files are physically stored; generating, by the profile engine, a multi-dimensional graph of a plurality of n-dimensions where each dimension of the n-dimensions corresponds with a respective attribute of the attributes; generating, by the profile engine, a bell curve along the n-dimensions of the multi-dimensional graph; determining, by the profile engine, which ones of the attributes impact the decision model more than a threshold amount according to a comparison of the attributes against the generated bell curve at a specified value for each of the attributes; identifying, by unsupervised machine learning, clusters of variable K to each of the attributes processed by K-means clustering that assigns each of the clusters to each of a plurality of K groups based on feature weights; positioning, by the profile engine on the multi-dimensional graph, the identified clusters, along a first axis comprising the n-dimensions and, along a second axis representing the feature weights, based on how much weight to give to each of the attributes along each of the n-dimensions; computing, by the profile engine, a distribution of each graph of the multi-dimensional graph along the first and second axes indicating how statistically reliable each of the n-dimensions are in determining predictable patterns of characterization for each of the identified clusters in response to respective combinations of the n-dimensions; recalibrating, by the profile engine, the feature weights of the attributes for the decision model, as the profile engine receives more data for the attributes according to a continuous decrease in variance between observed and expected outcomes of the decision model to improve accuracy over time of the weighing of each of the attributes within each of the n-dimensions; and generating, by the profile engine, along dimension lines of the multi-dimensional graph, a plurality of maximum nodes connected to a plurality of edges of a maximum value for each dimension of the n-dimensions along the dimension lines, indicating which particular dimension of the n-dimensions is most impactful in the decision model, according to said particular dimension having a higher value of its node up to the maximum nodes.
7. The system of claim 6, wherein the decision model corresponds to generating a financing structure.
8. The system of claim 6, where the attributes include: a final sale price of a specified product, a breakdown of a sale price, an amount of down payment being provided, condition of the specified product, whether the specified product is new or used, and a type of the specified product.
9. The system of claim 6, wherein the n-dimensions include: a percentage of backend value as compared to a total price of a specified product, frequency of counteroffers presented by sellers, weather on a day of a sale, an age of the specified product, and a color of the specific product.
10. The system of claim 6, wherein the profile engine is a dealer profile engine.
11. A non-transitory computer-readable medium storing instructions that when executed by one or more processors allow: concurrently executing, by a plurality of hypervisors, multiple instances of a plurality of operating systems to provide hardware virtualization; performing block virtualization, by a virtualized storage component of the hardware virtualization, that separates logical storage from physical storage to allow attributes of a data store to be accessed without regard to a heterogeneous structure; receiving, at a profile engine, the attributes accessed from the data store; performing file virtualization, using a virtualized storage component of the hardware virtualization, to eliminate dependencies between the accessed attributes at a file-level and a location where files

are physically stored; generating, by the profile engine, a multi-dimensional graph of a plurality of n-dimensions where each dimension of the n-dimensions corresponds with a respective attribute of the attributes; generating, by the profile engine, a bell curve along the n-dimensions of the multi-dimensional graph; determining, by the profile engine, which ones of the attributes impact a decision model more than a threshold amount according to a comparison of the attributes against the generated bell curve at a specified value for each of the attributes; identifying, by unsupervised machine learning, clusters of variable K to each of the attributes processed by K-means clustering that assigns each of the clusters to each of a plurality of K groups based on feature weights; positioning, by the profile engine on the multi-dimensional graph, the identified clusters, along a first axis comprising the n-dimensions and, along a second axis representing the feature weights, based on how much weight to give to each of the attributes along each of the n-dimensions; computing, by the profile engine, a distribution of each graph of the multi-dimensional graph along the first and second axes indicating how statistically reliable each of the n-dimensions are in determining predictable patterns of characterization for each of the identified clusters in response to respective combinations of the n-dimensions; recalibrating, by the profile engine, the feature weights of the attributes for the decision model, as the profile engine receives more data for the attributes according to a continuous decrease in variance between observed and expected outcomes of the decision model to improve accuracy over time of weighing of each of the attributes within each of the n-dimensions; and generating, by the profile engine, along dimension lines of the multi-dimensional graph, a plurality of maximum nodes connected to a plurality of edges of a maximum value for each dimension of the n-dimensions along the dimension lines, indicating which particular dimension of the n-dimensions is most impactful in the decision model, according to said particular dimension having a higher value of its node up to the maximum nodes.

12. The non-transitory computer-readable medium of claim 11, wherein the decision model corresponds to generating a financing structure.

13. The non-transitory computer-readable medium of claim 11, where the attributes include: a final sale price of a specified product, a breakdown of a sale price, an amount of down payment being provided, condition of the specified product, whether the specified product is new or used, and a type of the specified product.

14. The non-transitory computer-readable medium of claim 11, wherein the n-dimensions include: a percentage of backend value as compared to a total price of a specified product, frequency of counteroffers presented by sellers, weather on a day of a sale, an age of the specified product, and a color of the specific product.

15. The non-transitory computer-readable medium of claim 11, wherein the profile engine is a dealer profile engine.